

		Consortium for development of 25MW offshore wind turbine reference model
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25 MW Offshore Wind Turbine Reference Model

Data Sheet

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Document Number		Revision Number	A
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List of Alterations:

Revision	Date of Release	Modified Pages	Modification
A	2025	-	Modified by the institution(Kunsan univ,)

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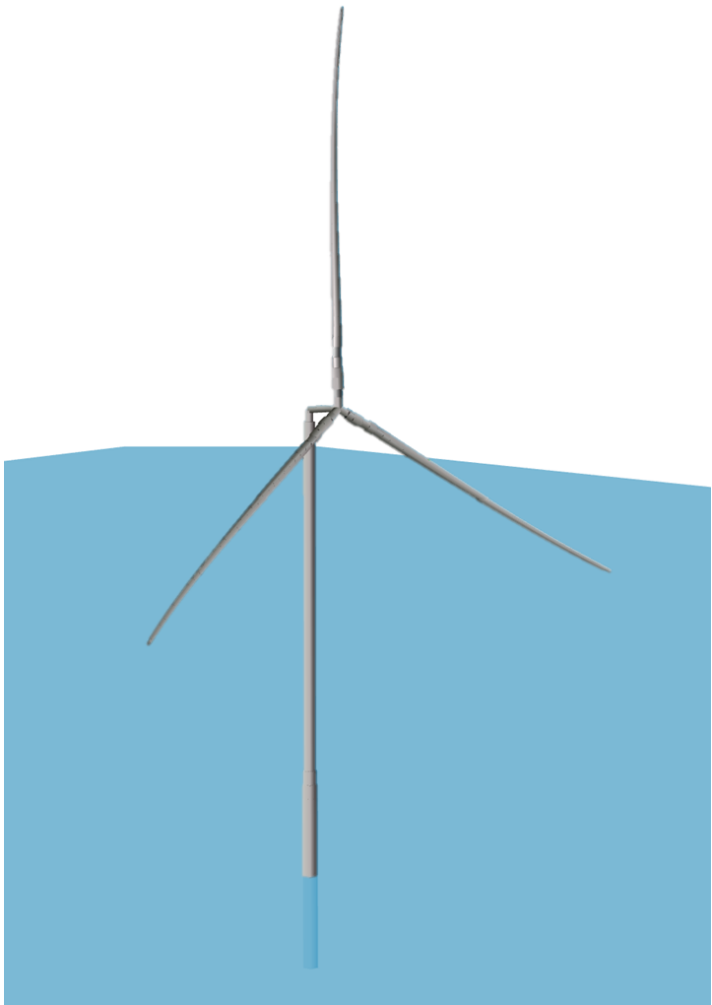
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1 General

This document provides a general technical overview of the 25MW offshore wind turbine reference model.

The reference model is a three-bladed upwind turbine collective pitch, direct drive permanent synchronous generator.



25MW Offshore Wind Turbine Reference Model

The turbine is designed to suit the conditions of the southern domestic sea in South Korea.

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2 Technical data

2.1 General data

Rotor diameter 315m
 Number of rotor blade 3
 Rated electrical power..... 25,000kW

Rotor speed range 2.0 – 5.95rpm
 Nominal rotor speed n_r 5.95rpm
 Nominal tip speed (at n_r) 98.14m/s
 Rated wind speed v_r 10.4m/s
 Optimum tip speed ratio 9.0
 Design type class 1B, T

Cut-in wind speed v_{in} 3m/s
 Cut-out wind speed v_{out} 25m/s
 Design lifetime 25 Years

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2.2 Rotor

Rotor position related to tower upwind

Rotor turning direction (view from roto to nacelle) clockwise

Hub mass 131.0 t

Blade 1ea mass.....140.0 t

Rotor mass (Hub and Blade 3ea).....551.0 t

2.3 Nacelle

Hub diameter 8.4 m

Nacelle length 7.6 m

Nacelle width.....5.85 m

Nacelle height.....6.53 m

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2.4 Generator and Converter

Rated power (electrical output)	26,900 kW
Rated voltage	3,300 V
Speed range.....	2 – 5.95 rpm
Rated speed	5.95 rpm
Maximum overspeed.....	7.14 rpm
Total generator mass.....	approx. 530 t
Rated power output converter	25,600 kW
Converter mass.....	approx. 23.0 t

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2.5 Tower

TypeTubular steel tower

Top diameter..... 6.0 m

Tower weight1,664 t

Tower height..... 167 m

Hub height 195 m

Bottom diameter.....10.5 m

Number of sections1

2.6 Monopile

TypeTubular steel tower

Top diameter.....10.5 m

Monopile height71.5 m

Monopile height (under water)49.0 m

Number of sections1

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3 Design wind speeds at hub height

3.1 1B, T

Annual average wind speed..... 8.6 m/s
1 year extreme wind speed 34.3 m/s
50 year extreme wind speed 50.8 m/s